Survival-Aware Scheduling for Autologous CAR-T Supply Chains
Methods

3 Methods

3.1 Overview of the modelling framework

Autologous CAR-T therapy couples a manufacturing supply chain to a deteriorating patient. Each therapy is patient-specific and non-substitutable: the cells collected from a patient can be manufactured only for that patient, so a delay in the network is not a delay in inventory but a delay in treatment, borne by a patient whose probability of surviving to infusion is falling. Conventional supply-chain models treat time through a deadline—a maximum admissible turnaround—which is a binary and tier-blind representation of a hazard that is in fact continuous and tier-specific. The framework developed here replaces the deadline with an explicit survival model, and in doing so makes clinical prioritisation an *output* of the optimisation rather than an input rule.

The framework has two layers, solved in sequence (Figure 1).

The **strategic layer** (§3.3) is a mixed-integer linear program that extends the patient-centric supply-chain design platform of Triantafyllou et al. [1] in two ways. First, it introduces a genuine manufacturing queue: the start of manufacturing becomes a decision variable constrained by material arrival and by concurrent facility capacity, so a job may be *held*. Second, it prices that hold clinically, through a patient-level survival function embedded exactly in the objective. The layer decides facility locations, patient–facility assignment, transport modes, and a nominal schedule.

The **operational layer** (§3.5–§3.6) freezes the strategic decisions and asks the question the deterministic model cannot: what happens when manufacturing actually fails? A daily discrete-event simulation advances the clock over the planning horizon, realises manufacturing yield as a Bernoulli outcome at release testing, evolves each patient's survival with the accumulated wait, and at every epoch at which a manufacturing slot frees calls a small look-ahead optimisation that decides who starts next on the *observed* state. Manufacturing failure triggers a re-collection recourse gated on projected survival.

This separation is deliberate and reflects the decision horizons of the real system: facilities and contracts are committed once, over years; sequencing decisions are made daily, under information that did not exist when the network was designed. It also yields a clean identification strategy. Because the network is frozen and shared across every policy we compare, and because manufacturing failures are generated by common random numbers keyed to the patient and attempt rather than to call order (§3.9), any difference in outcome between two policies is attributable to sequencing alone—not to a different network, a different cost structure, or a luckier draw.

3.2 Notation

Sets: patients $p \in P$ with risk tier $u(p) \in \{H, M, L\}$; manufacturing sites $m \in M$; leukapheresis sites $c \in C$; hospitals $h \in H$; transport modes $j \in J$; periods $t \in T = \{1, \ldots, 130\}$; and

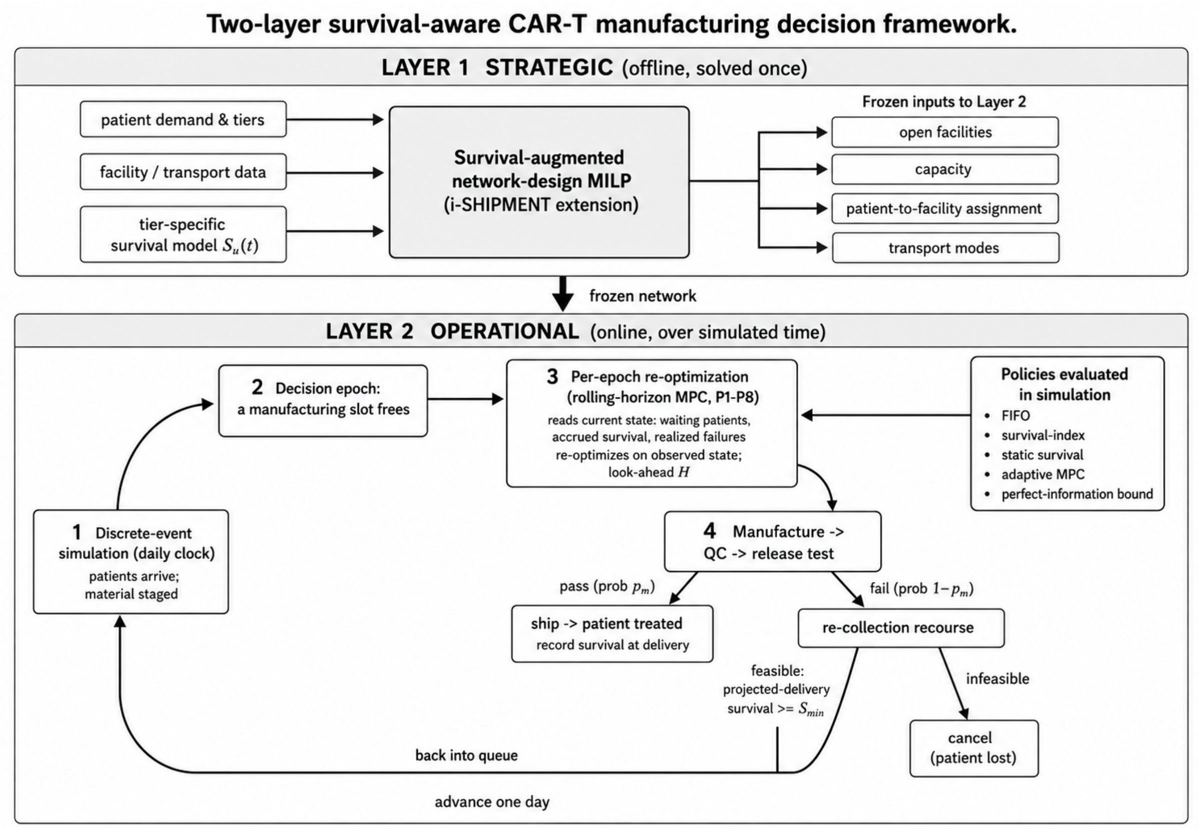

Figure 1: **Two-layer survival-aware scheduling framework.** The strategic layer is solved once and fixes the network—open facilities, capacity, patient–facility assignment and transport modes. The operational layer executes on that frozen network: a daily discrete-event simulation realises manufacturing yield, and at each epoch at which a slot frees the per-epoch problem re-optimises on the observed state over a look-ahead window H, implementing only the immediate start. A failed batch returns its patient to the queue subject to the re-collection gate, or is cancelled. All five policies enter at the per-epoch decision point and differ only there.

$\Theta \subset C \times H$ the set of co-located leukapheresis–hospital pairs.

Parameters retained from the base model: capital and variable facility cost CIM_m, CVM_m; concurrent capacity FCAP_m; unit transport costs $U1_{cmj}$, $U3_{mhj}$; transport times $\mathrm{TT1}_j$, $\mathrm{TT3}_j$; stage durations T_{LS}, T_{MFE}, T_{QC}; per-therapy material and release-testing costs C_{mat}, C_{QC}; the arrival incidence INC_{pct}; the flow bounds FMIN, FMAX; and the maximum turnaround ND.

Parameters introduced here: the tier reference mortality w_u and the induced hazard HR_u; the survival scale η and shape γ, and sensitivity κ; the clinical weight ρ_u; the cost–survival exchange rate α; the manufacturing yield p_m; the look-ahead horizon H; the re-collection floor $S_{\min}$; the maximum number of remakes K_{remake}; and the re-leukapheresis cost ρ_{leuk}.

Decision variables retained: facility opening $E1_m \in \{0,1\}$; link activation $X1_{cm}, X2_{mh} \in \{0,1\}$; patient routing $Y1_{pcmjt}$, $Y2_{pmhjt} \in \{0,1\}$; the material flow variables LSA, LSR, MSO, OUTC, OUTM, INM, DURV, FTD, INH; and the timing variables STT_p, CTT_p, TRT_p. Variables introduced here: the survival rate $S_p \in [0,1]$; the queue delay $\mathrm{HOLD}_p \geq 0$; the material arrival A_{pmt}; and the survival indicators $\delta_{p,d} \in \{0,1\}$.

3.3 Strategic layer: survival-aware network and queue design

3.3.1 Relationship to the base model

The strategic layer is built on the supply-chain design platform of Triantafyllou et al. [1], and it is worth stating precisely which components are theirs and which are introduced here.

Retained without modification. The network and material-flow structure is theirs: facility siting and link activation, the patient-routing variables, the time-indexed material balances that carry each patient's material through leukapheresis, inbound transport, manufacturing, release testing and outbound transport, the facility-capacity accounting, the co-location requirement between collection site and treating hospital, and the cost structure comprising amortised facility cost, transport cost and per-therapy material and release-testing cost. In the formulation that follows this is Equations (2)–(26), together with the first three terms of the objective (1). The instance data—candidate sites, capacities, transport times and unit costs—is likewise theirs.

Introduced in this work. Four things are new. First, manufacturing start becomes a decision rather than being identified with material arrival, which creates the queue and with it the possibility of a deliberate hold, Equations (27)–(30). Second, patient survival enters the model explicitly, as the final term of (1) and its exact linearisation (31)–(34). Third, the maximum-turnaround constraint is relaxed from the binding timeliness mechanism to a slack outer bound (35), for the reason given in §3.3.5. Fourth, the entire operational layer—the per-epoch decision problem, the execution environment, the policies and the perfect-information benchmark—has no counterpart in the base model, which is deterministic and solved once.

3.3.2 Objective

The planner minimises supply-chain cost plus a monetised clinical loss,

$$\min\ Z \ =\ \sum_p \mathrm{CTM}_p \ +\ \sum_p \mathrm{TTC}_p \ +\ (C_{\mathrm{mat}} + C_{\mathrm{QC}})\,|P| \ +\ \alpha \sum_p \rho_{u(p)}\big(1 - S_p\big), \tag{1}$$

where CTM_p is the amortised facility cost and TTC_p the transport cost of patient p,

$$\mathrm{CTM}_p = \sum_m E1_m(\mathrm{CIM}_m + \mathrm{CVM}_m)\,|T|/|P|, \qquad \forall p, \tag{2}$$

$$\mathrm{TTC}_p = \sum_{c,m,j,t} Y1_{pcmjt}\, U1_{cmj} + \sum_{m,h,j,t} Y2_{pmhjt}\, U3_{mhj}, \qquad \forall p. \tag{3}$$

The final term of (1) is the contribution of this work. It carries units of dollars: α is an explicit value of a statistical life and ρ_u a dimensionless triage weight, so $\alpha\rho_u$ is the per-tier dollar valuation of an avoided death. We stress that (1) contains **no priority rule, no tier ordering and no reserved capacity**. Any prioritisation that emerges does so because S_p falls faster for higher-hazard tiers, which is a property of the calibration in §3.4, not of the constraint set.

3.3.3 Material balance, capacity and network structure

Equations (4)–(12) are the time-indexed material balances of the base model, propagating each patient's material from collection through leukapheresis, inbound transport, manufacturing, release testing and outbound transport:

$$\begin{aligned}
\mathrm{INC}_{pct} &= \mathrm{OUTC}_{p,c,t+T_{\mathrm{LS}}}, && \forall p,c,t, && (4)\\
\mathrm{OUTC}_{pct} &= \sum_{m,j} \mathrm{LSR}_{pcmjt}, && \forall p,c,t, && (5)\\
\mathrm{LSR}_{pcmjt} &= \mathrm{LSA}_{p,c,m,j,t+\mathrm{TT1}_j}, && \forall p,c,m,j,t, && (6)\\
A_{pmt} &= \sum_{c,j} \mathrm{LSA}_{pcmjt}, && \forall p,m,t, && (7)\\
\mathrm{INM}_{pmt} &= \mathrm{OUTM}_{p,m,t+T_{\mathrm{MFE}}}, && \forall p,m,t, && (8)\\
\mathrm{OUTM}_{pmt} &= \sum_{h,j} \mathrm{MSO}_{p,m,h,j,t+T_{\mathrm{QC}}}, && \forall p,m,t, && (9)\\
\mathrm{MSO}_{pmhjt} &= \mathrm{FTD}_{p,m,h,j,t+\mathrm{TT3}_j}, && \forall p,m,h,j,t, && (10)\\
\mathrm{INH}_{pht} &= \sum_{m,j} \mathrm{FTD}_{pmhjt}, && \forall p,h,t, && (11)\\
\mathrm{DURV}_{pmt} &= \sum_{1<\tau\le t} \big(\mathrm{INM}_{p,m,\tau-1} - \mathrm{OUTM}_{p,m,\tau}\big) + \mathrm{OUTM}_{pmt}, && \forall p,m,t. && (12)
\end{aligned}$$

Equation (7) is where this model departs from its predecessor. In the base model the manufacturing start is *identified* with material arrival: a job begins the day its material reaches the site. Here A_{pmt} records only the arrival, and the start INM is released as a decision, governed by the queue (27)–(30) below.

Concurrent manufacturing at each site is tracked and capped by

$$\begin{aligned}
\mathrm{RATIO}_{mt} &= \sum_{p} \mathrm{DURV}_{pmt}/\mathrm{FCAP}_m, && \forall m,t, && (13)\\
\mathrm{CAP}_{mt} &= \mathrm{FCAP}_m - \sum_{p} \sum_{t-T_{\mathrm{MFE}}\le\tau<t} \mathrm{INM}_{p,m,\tau}, && \forall m,t, && (14)\\
\sum_{p} \mathrm{INM}_{pmt} - \sum_{p} \mathrm{OUTM}_{pmt} &\le \mathrm{CAP}_{mt}, && \forall m,t. && (15)
\end{aligned}$$

The network structure—at most two open facilities, links only to open facilities, exactly one journey per leg, flow permitted only on activated journeys, and delivery to the hospital co-located with the collection site—is carried over unchanged:

$$\sum_m E1_m \le 2, \tag{16}$$

$$X1_{cm} \le E1_m, \quad X2_{mh} \le E1_m, \qquad \forall c, m, h, \tag{17}$$

$$Y1_{pcmjt} \le X1_{cm}, \quad Y2_{pmhjt} \le X2_{mh}, \qquad \forall p, c, m, h, j, t, \tag{18}$$

$$\sum_{c,m,j,t} Y1_{pcmjt} = 1, \quad \sum_{m,h,j,t} Y2_{pmhjt} = 1, \qquad \forall p, \tag{19}$$

$$\sum_{p,h,t} \mathrm{INH}_{pht} \le |P|, \tag{20}$$

$$Y1_{pcmjt}\,\mathrm{FMIN} \le \mathrm{LSR}_{pcmjt} \le Y1_{pcmjt}\,\mathrm{FMAX}, \qquad \forall p, c, m, j, t, \tag{21}$$

$$Y2_{pmhjt}\,\mathrm{FMIN} \le \mathrm{MSO}_{pmhjt} \le Y2_{pmhjt}\,\mathrm{FMAX}, \qquad \forall p, m, h, j, t, \tag{22}$$

$$\sum_{m,j,t} Y2_{pmhjt} = \sum_t \mathrm{INC}_{pct}, \qquad \forall p,\ \forall (c,h) \in \Theta. \tag{23}$$

Timing follows

$$\mathrm{STT}_p = \sum_{c,t} t\,\mathrm{INC}_{pct}, \quad \mathrm{CTT}_p = \sum_{h,t} t\,\mathrm{INH}_{pht}, \qquad \forall p, \tag{24}$$

$$\mathrm{TRT}_p = \mathrm{CTT}_p - \mathrm{STT}_p, \quad \mathrm{STT}_p \le \mathrm{CTT}_p, \qquad \forall p, \tag{25}$$

$$\mathrm{ATRT} = \tfrac{1}{|P|} \sum_p \mathrm{TRT}_p, \tag{26}$$

where TRT_p is the turnaround time of patient p, measured from collection to infusion, and ATRT its cohort average.

3.3.4 The manufacturing queue

Replacing the fixed start with a queue requires four constraints:

$$\sum_{\tau \le t} \mathrm{INM}_{pm\tau} \le \sum_{\tau \le t} A_{pm\tau}, \qquad \forall p, m, t, \tag{27}$$

$$\sum_t \mathrm{INM}_{pmt} = \sum_t A_{pmt}, \qquad \forall p, m, \tag{28}$$

$$\sum_p \mathrm{DURV}_{pmt} \le \mathrm{FCAP}_m, \qquad \forall m, t, \tag{29}$$

$$\mathrm{HOLD}_p = \sum_{m,t} t\left(\mathrm{INM}_{pmt} - A_{pmt}\right) \ \ge\ 0, \qquad \forall p. \tag{30}$$

Constraint (27) forbids a job from starting before its material has arrived; (28) requires each routed material to be started exactly once; (29) caps concurrent manufacturing at each site; and (30) defines the hold, the delay between arrival at the manufacturing site and the start of manufacturing.

The hold is the **only new degree of freedom in the model**, and this is what makes the design identifiable. Capacity (29) forces some patients to wait; the objective (1) chooses *which* patients wait. Because the downstream balances (8)–(11) shift with the chosen start, TRT_p—and hence S_p—absorb the wait automatically, without any additional linking constraint.

3.3.5 Exact linearisation of survival

The survival model is

$$S_p = \exp\Big(-\kappa \cdot \mathrm{HR}_{u(p)} \cdot \left(\mathrm{TRT}_p/\eta\right)^{\gamma}\Big), \qquad \forall p, \tag{31}$$

which is nonlinear in TRT_p. Two features of the problem make an *exact* linearisation available rather than an approximation. First, every duration in the network is integral, so TRT_p is integer-valued. Second, TRT_p is bounded below by the fastest feasible route and above by the turnaround cap,

$$T_{\min} = T_{\mathrm{LS}} + \min_j \mathrm{TT1}_j + T_{\mathrm{MFE}} + T_{\mathrm{QC}} + \min_j \mathrm{TT3}_j = 17 \text{ days}, \qquad T_{\max} = \mathrm{ND} = 42 \text{ days},$$

so TRT_p is supported on the 26 integers $\{17, \ldots, 42\}$. We therefore introduce binary indicators $\delta_{p,d}$, $d \in \{17, \ldots, 42\}$, and impose

$$\sum_d \delta_{p,d} = 1, \qquad \forall p, \tag{32}$$

$$\mathrm{TRT}_p = \sum_d d\, \delta_{p,d}, \qquad \forall p, \tag{33}$$

$$S_p = \sum_d \sigma_{d,u(p)}\, \delta_{p,d}, \qquad \forall p, \tag{34}$$

with $\sigma_{d,u} = S_u(d)$ precomputed from (31). Because the support of TRT_p is exactly the breakpoint set, (32)–(34) is not a piecewise-linear approximation: it reproduces (31) exactly at every feasible point. This is a strengthening of the SOS2 λ-method used in the original model specification, which the δ-formulation dominates here precisely because integrality removes the need for interpolation between breakpoints. We report the implemented formulation rather than the specified one, and note that the two coincide in value.

Finally,

$$\mathrm{TRT}_p \leq \mathrm{ND}, \qquad \forall p, \tag{35}$$

with ND relaxed from 18 to 42 days. This relaxation is essential and deserves comment: under $\mathrm{ND} = 18$ the turnaround is pinned to $\{17, 18\}$ and the survival term of (1) has almost no support to act on, so the model *cannot* express a prioritisation preference. Relaxing ND converts the deadline from the binding timeliness mechanism into a slack outer bound, and lets survival—a continuous, tier-specific quantity—drive timeliness instead. The relaxation does not weaken the clinical requirement; it relocates it from a constraint to the objective, where it can be traded against cost at a stated exchange rate.

3.3.6 Exact reformulation and valid inequalities

Three reductions make the model tractable at the scales studied without altering its feasible set or optimal value.

Index reduction. In every feasible solution the routing variables collapse: constraints (19), (21) and (23) force the single active $Y1$ of patient p onto its own collection site and onto the single day $t = t_p^0 + T_{\mathrm{LS}}$, and force the single active $Y2$ onto the hospital co-located with that site. We therefore replace $Y1_{pcmjt}$ by $y1_{pmj}$ and $Y2_{pmhjt}$ by $y2_{pmj}$, eliminating two index dimensions. Likewise (13)–(15) with (12) reduce algebraically to a rolling T_{MFE}-day window on manufacturing starts, $\sum_p \sum_{\tau=t-T_{\mathrm{MFE}}+1}^{t} \mathrm{INM}_{pm\tau} \leq \mathrm{FCAP}_m$. Equivalence of the reduced and full-index formulations is verified numerically at the 50-patient scale rather than asserted.

Dominance cuts. Sites m_4, m_5, m_6 share the capital cost, variable cost and capacity of m_1, m_2, m_3 respectively but carry weakly higher transport costs from every collection site. Any solution opening the dominated site without its counterpart can be relabelled onto the counterpart at no greater cost and with identical timing, so $E1_{m_4} \leq E1_{m_1}$, $E1_{m_5} \leq E1_{m_2}$, $E1_{m_6} \leq E1_{m_3}$ are valid and objective-preserving.

Symmetry breaking. Patients sharing a collection site, a risk tier and an arrival day are fully interchangeable—identical costs, arrival dynamics and survival curves—so requiring their start days to be non-decreasing removes a symmetry group without cutting off any objective value.

Aggregate throughput cuts. Constraint (29) admits at most FCAP_m starts in any T_{MFE} consecutive days. Partitioning the start-window union into disjoint T_{MFE}-day blocks and applying the argument to every prefix yields a family of cuts of the form $k \sum_m \mathrm{FCAP}_m E1_m \geq$ (number of patients whose start window ends by T). These are implied by (29) and therefore change neither the feasible set nor the optimum; they tighten the root relaxation, which is what makes the largest instances provable.

3.4 Survival calibration

The survival model is calibrated to a reference six-week mortality rather than to a hazard ratio, which makes every parameter clinically interpretable and identifiable from published outcome data.

Let w_u denote the probability that a tier-u patient dies within $T_{\mathrm{ref}} = 42$ days without treatment, so $S_u(42) = 1 - w_u$. Setting $\eta = T_{\mathrm{ref}}$ and $\kappa = 1$ fixes the scale, and the hazard follows as $\mathrm{HR}_u = -\ln(1 - w_u)$, which is *derived* rather than assumed. With $w = \{H\colon 0.15,\ M\colon 0.05,\ L\colon 0.02\}$ this yields $\mathrm{HR} = \{0.163, 0.051, 0.020\}$, a ratio of roughly $8 : 2.5 : 1$.

We fix the shape $\gamma = 1$, giving a constant baseline hazard and the closed form

$$S_u(t) = (1 - w_u)^{t/42}. \tag{36}$$

This is the conservative choice, and the reason matters for the credibility of the results. Under $\gamma = 1$ a saved day is worth the same at every point in the patient's wait; any prioritisation benefit the model finds therefore cannot be an artefact of an accelerating hazard curve that mechanically rewards serving the sickest first. Values $\gamma = 1.5$ and 2 are retained only as a robustness sweep. Survival is monotone for every $\gamma > 0$, so the qualitative structure is unaffected.

At the minimum turnaround of 17 days the high-risk tier already sits at $S_H = 0.936$ against $S_L = 0.992$—a 5.6-point gap that widens to 13.0 points at 42 days. That gap is the entire room the objective has to work with, and it is absent from the base model, in which turnaround is pinned to 17–18 days.

3.5 Operational layer: the per-epoch decision problem

With the strategic layer frozen, a decision epoch occurs whenever a manufacturing slot frees at an open site. At epoch t the observed state consists of the ready set W_t of patients whose material is staged at their assigned site; for each $i \in W_t$, its tier $u(i)$, its accrued wait a_i, a failed-once indicator φ_i and its frozen site $m(i)$; and the occupancy $\mathrm{busy}_m(\tau)$ contributed by in-progress jobs. The only decision is

$$x_{i,\tau} \in \{0,1\} : \text{ start patient } i\text{'s manufacturing on day } \tau \in [t, t+H] \text{ at } m(i).$$

Survival is evaluated from the observed state, not from a nominal plan:

$$\mathrm{elapsed}_i(\tau) = a_i + (\tau - t) + T_{\mathrm{MFE}} + T_{\mathrm{QC}} + \mathrm{TT3}_{m(i)}, \qquad \forall i \in W_t,\ \tau, \tag{37}$$

$$S_i(\tau) = \left(1 - w_{u(i)}\right)^{\mathrm{elapsed}_i(\tau)/42}, \qquad \forall i \in W_t,\ \tau. \tag{38}$$

Because the survival clock runs continuously from the *first* collection, $a_i = t - t_i^0$, and (37) simplifies to $\mathrm{elapsed}_i(\tau) = \tau - t_i^0 + T_{\mathrm{MFE}} + T_{\mathrm{QC}} + \mathrm{TT3}_i$. A remade patient therefore carries the deterioration of every earlier attempt, which is the property that makes repeated failure clinically costly rather than merely expensive.

The epoch problem is

$$\min \sum_{i \in W_t} \rho_{u(i)} \big(1 - \hat{S}_i\big), \tag{39}$$

with effective survival

$$\hat{S}_i = \sum_{\tau} S_i(\tau)\, x_{i,\tau} + d_i \Big(1 - \sum_{\tau} x_{i,\tau}\Big), \qquad d_i = S_i(t+H), \tag{40}$$

subject to

$$\sum_{\tau} x_{i,\tau} \le 1, \qquad \forall i \in W_t, \tag{41}$$

$$\sum_{i:\, m(i)=m} \; \sum_{\tau - T_{\mathrm{MFE}} < \tau' \le \tau} x_{i,\tau'} \le \mathrm{FCAP}_m - \mathrm{busy}_m(\tau), \qquad \forall m,\ \tau \in [t, t+H], \tag{42}$$

$$x_{i,\tau} = 0, \qquad \text{for } \tau < t. \tag{43}$$

The continuation term d_i in (40) is what prevents (39) from degenerating into a myopic rule: deferring a patient past the window is charged at the survival that deferral would cost, so the model weighs serving a patient now against the damage of not serving them at all within the horizon. The frozen operational cost c_i^{op} is inert on a frozen network—it is a per-patient constant and every started patient is started exactly once—so it does not enter the argmin and is not monetised into (39); α appears only in the strategic objective (1).

Two structural properties are worth stating. First, because $m(i)$ is frozen, the patient set partitions across facilities and (42) never couples two of them: the epoch problem **decomposes exactly by facility**, so solving it per site is solving it globally. Second, only the immediate start ($\tau = t$) is implemented before the simulation advances and the problem is re-solved; the remainder of the look-ahead plan is discarded, in the standard receding-horizon manner.

3.5.1 The interpretable index rule

In the special case of a single free slot, $H = 0$ and no operational cost, (39)–(43) collapse to a closed form:

$$i^* = \arg\max_{i \in W_t} \; \rho_{u(i)} \big[\, S_i(t) - S_i(t+1) \,\big]. \tag{44}$$

Serve whoever's ρ-weighted survival falls most from one further day of waiting. This is a Whittle-type index: it requires no solver, is inspectable by a clinician, and is the natural rule to deploy. We evaluate it as a policy in its own right, and its relationship to the full look-ahead model is an empirical question we report rather than assume.

3.6 Execution environment: discrete-event simulation

The simulation advances a daily clock $t = 1, \ldots, T_{\mathrm{horizon}} = 130$ over the frozen network. Within each day, events are processed in the following order.

1. **Arrivals.** Patients collected on day t enter the system; after $T_{\mathrm{LS}} + \mathrm{TT1}$ days their material is staged and they join W.

2. **Completions and yield.** A job started on day s holds its slot for T_{MFE} days and frees it on day $s + T_{\mathrm{MFE}}$. T_{QC} days later it is release-tested: it passes with probability p_m and is delivered TT3 days later, at which point the patient is treated and their realised survival recorded; it fails with probability $1 - p_m$, returning the patient to W with $\varphi_i = 1$ and a larger accrued wait.

3. **Health update.** Accrued waits advance; because the clock is continuous from first collection this is implicit in $t - t_i^0$.

4. **Decision.** For each site with a free slot and ready patients, the policy is called and the chosen jobs start.

5. **Losses.** A patient leaves the system unserved only through the recourse rules of §3.6.1.

Manufacturing yield is the sole stochastic element. Arrivals are the fixed instance arrivals and health evolves deterministically given the wait, so randomness enters at exactly one point and its effect on outcomes is identifiable. Expected loss is accumulated as the probability sum $\sum_i (1 - S_i)$ on each replication's realised timeline, with $S = 0$ for a patient never treated; this is a lower-variance estimator than Bernoulli survival draws and is exact given the timeline.

3.6.1 Failure recourse with re-collection

On a manufacturing failure the patient requires a *fresh* leukapheresis. The recourse is deterministic, consistent with deterministic health evolution.

Feasibility. Re-collection is admissible if and only if projected survival at the remake's delivery still clears the floor,

$$\hat{S}_i(\text{delivery}) = \big(1 - w_{u(i)}\big)^{\left[a_i + (T_{\text{LS}} + \text{TT1}) + wait_i + T_{\text{MFE}} + T_{\text{QC}} + \text{TT3}\right]/42} \geq S_{\min}, \tag{45}$$

where $wait_i$ is the days until the next free slot at $m(i)$, read from the same occupancy $\text{busy}_m(\cdot)$ that (42) uses. Because the floor is evaluated on *projected* rather than current survival, sicker and later patients cross it first, reproducing tier-dependent re-collection eligibility without introducing a second stochastic primitive.

If feasible, the remake restarts from collection: $T_{\text{LS}} + \text{TT1}$ days are added before manufacturing can resume, the accrued wait grows accordingly, and ρ_{leuk} is charged. The patient re-enters W as a $\varphi = 1$, lower-survival competitor.

If infeasible, or once K_{remake} remakes are exhausted, the patient is cancelled and carries the full clinical loss.

There is no calendar-based removal of any kind—no raw-wait eligibility cutoff. A patient still queueing keeps queueing and is credited with whatever survival their eventual delivery earns, including deliveries falling beyond the reporting horizon. §3.10 documents why this choice is material.

3.7 Policies compared

Five policies are executed on the identical frozen network, with identical realised failures.

The four online policies are strictly non-idling: a free slot is never left empty while a ready patient waits. This is imposed as an equality on the immediate starts,

$$\sum_{i \in W_t} x_{i,t} = \min\big(\text{FCAP}_m - \text{busy}_m(t),\ |W_t|\big), \qquad \forall m, \tag{46}$$

so that policies differ in *whom* they select and never in *how much* capacity they use—without (46), a policy could appear to reduce clinical loss simply by withholding capacity.

The design isolates two distinct effects. `fifo` versus the survival-aware policies identifies the value of survival awareness. `static_survival` versus `adaptive_mpc` identifies the value of *adaptivity* specifically, since both optimise the same clinical objective and differ only in whether they re-solve on observed state.

Table 1: The five operating policies.

Policy	Rule
`fifo`	Serve in arrival order; no objective. The do-nothing reference.
`survival_index`	The closed-form index (44).
`static_survival`	The survival schedule optimised once up front—the strategic start order—dispatched greedily and never re-solved. Remakes join the back of the order.
`adaptive_mpc`	The full per-epoch model (39)–(43), re-solved at every epoch on the observed state.
`best_achievable`	A perfect-information benchmark; see §3.8.

3.8 The perfect-information benchmark

A policy comparison without an attainable floor cannot distinguish a good policy from an easy instance. We therefore compute, for every replication, the optimal schedule *given the realised failures.*

Let $f_{i,k} \in \{0,1\}$ indicate that attempt k of patient i fails under the replication's random draws—known to this model, unknown to every online policy. Binary variables $x_{i,k,\tau}$ start attempt k of patient i on day τ. Attempt $k+1$ may be run only if attempt k was run and failed, and no earlier than that attempt's release plus a re-collection:

$$\sum_{\tau} x_{i,k+1,\tau} \le \sum_{\tau} x_{i,k,\tau}, \qquad \forall i,k, \tag{47}$$

$$\sum_{\tau} \tau\, x_{i,k+1,\tau} \ge \sum_{\tau} \tau\, x_{i,k,\tau} + T_{\mathrm{MFE}} + T_{\mathrm{QC}} + T_{\mathrm{LS}} + \mathrm{TT1} - M\Big(1 - \sum_{\tau} x_{i,k+1,\tau}\Big), \quad \forall i,k. \tag{48}$$

These are imposed subject to the same rolling-window capacity the simulation enforces and the same $S_{\min}$ admissibility (45). The objective maximises realised weighted survival, in which only a successful attempt delivers anything:

$$\max \sum_{i} \sum_{k:\, f_{i,k}=0} \sum_{\tau} \rho_{u(i)}\, S_{u(i)}\big(\tau + T_{\mathrm{MFE}} + T_{\mathrm{QC}} + \mathrm{TT3}_i - t_i^0\big)\, x_{i,k,\tau}. \tag{49}$$

Three points of interpretation are essential. First, the resulting value is a valid lower bound on the **ρ-weighted clinical loss** $\sum_p \rho_{u(p)}(1-S_p)$—the clinical-loss term of objective (1)—and *not* on unweighted patient counts; a policy can and does fall below it on unweighted totals without contradiction, and we report it only against the quantity it bounds. Second, this benchmark is deliberately exempt from the non-idling requirement (46): deliberately holding a slot open for a sicker patient arriving tomorrow is part of what perfect information buys, and forcing it to be work-conserving would produce something other than a bound. Third, declining an attempt is itself one of the decisions foresight permits, since capacity spent on a batch known to fail is capacity denied to another patient.

The benchmark is executed through the same simulation and accounting code as every other policy, so that reported costs and metrics are constructed identically and remain comparable.

3.9 Experimental design

Instances. All experiments use the published network data of the base model—four collection sites, six candidate manufacturing sites with FCAP $\in \{4, 31, 10, 4, 31, 10\}$, two transport modes—with cohorts built by tiling the 50-patient base cohort. Tier mix (30/40/30 H/M/L) and the collection-site distribution are inherited patient by patient, and arrival days are rescaled into the admissible window so that arrival *density*, and hence congestion, grows with cohort size. Experiments run at $N = 200$ (primary) and $N = 100$ (confirmation); the 50-patient cohort is used for schedule-level illustration.

Replication and variance reduction. Each configuration is replicated over 30 yield seeds. The draw governing attempt k of patient i is a deterministic function of (seed, patient, attempt) rather than of call order, which delivers common random numbers in a strong form: the same batches fail under every policy, at every offered load, and nested across the failure-rate sweep. Policy contrasts are therefore paired, and the reported standard errors are of differences that share their noise.

Design of experiments. Five experiments address five distinct questions: the calibration itself (does a day of delay cost tiers differently?); the prioritisation mechanism (who is served first, and where does the waiting go?); congestion (when does survival awareness matter?), varied by compressing or expanding the arrival window on a fixed network so that offered load moves without confounding the network; the value of life (at what α does the network *design* change?), the only experiment that re-solves the strategic layer; the policy benchmark against the bound; and manufacturing failure rate, swept as a common $(1 - p)$ across all sites so the abscissa is interpretable.

Reported metrics. We restrict reporting to five quantities fixed in advance: expected high-risk patients lost (primary), total expected patients lost, total cost and cost per therapy, mean hold by tier, and the share of the `fifo`-to-`best_achievable` gap closed. Both the primary and the total metric are reported in every experiment, in both directions, so that a policy which improves one at the expense of the other cannot be presented selectively.

3.10 Modelling choices, limitations and reproducibility

Several choices bound the interpretation of the results and are stated here rather than in a concluding caveat.

The re-collection floor is weakly binding. In (45) the term $wait_i$ is read from in-progress occupancy, which by construction never looks more than $T_{\mathrm{MFE}} - 1 = 6$ days ahead. The floor $S_{\min} = 0.75$ therefore requires an accrued wait exceeding roughly 50 days for the high-risk tier—and far more for the others—before it can bind. In practice it binds only for policies whose rigid ordering allows individual patients to decay substantially. A specification in which $wait_i$ reflected queue position rather than in-progress occupancy would make the floor active across all policies; we retain the stated specification and report the consequence rather than tuning the rule to produce activity.

Removal versus decay. Because there is no calendar-based removal, expected loss is dominated by survival decay accumulated while waiting rather than by patients removed from the system. This is the intended semantics—the model prices delay, and delay is continuous—but it means "expected patients lost" is an expected-mortality quantity, not a count of censored patients.

Scope of the frozen network. Assignment and transport mode are inherited from the strategic solution and are not re-decided per epoch. Expediting an individual patient by upgrading transport mid-horizon is therefore outside the policy space, and the reported gains are attributable to sequencing alone. This understates what a fully flexible operational layer could achieve and is conservative with respect to our claims.

Horizon spillover. Deliveries falling beyond the 130-day reporting horizon are credited with their realised survival rather than discarded or zeroed. Spillover is below 5% at the operating point and is reported for every configuration; at the lowest offered load, where the arrival window is deliberately stretched beyond the horizon, it is substantially larger and is reported as such.

Computational implementation. All programs are implemented in Pyomo and solved with a commercial branch-and-cut solver, falling back to an open-source solver where licence limits preclude it. The strategic and perfect-information programs are solved to proven optimality at every scale reported. The per-epoch programs are small—at most a few hundred binary columns, since the ready queue at a single facility never exceeded 33 patients—which is what makes re-optimisation at every epoch practical rather than merely conceivable. Every replication records the random seed that generated it and the strategic solution it was executed against, so each reported quantity is reproducible from a stated draw.

References

[1] N. Triantafyllou et al. A digital platform for the design of patient-centric supply chains. *Scientific Reports*, 2022.